MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL-ALUMINUM, TYPE I, CLASS 2, COMPOSITION G

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Ni	Al	Others (1)	
		Each	Total
Bal	17.0 20.0	-	1.0 max.

(1) Determination not required for routine acceptance.

Particle Size/Distribution

+140 Mesh (+160 µm)	0.5 max.
+170 Mesh (+90 µm)	5.0 max.
+270 Mesh (+53 µm)	93.0 min.

Metallographic Inspection

Powder particles shall be capable of exhibiting a structure of an aluminum, spheroidal core essentially enclosed in a nickel shell.

Sprayed Coating Properties

Bond Strength	34.0 MPa (5000 psi) min.
Hardness	Rockwell 15T, 82 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 2% max.

AMSC N/A FSC 3439

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:31Z
Check the source to verify that this is the current version before use.

Deposit Efficiency and Deposition Rate

55% min for N_2/H_2 @ 9.75 lb/hr
40% min for A/H_2 @ 7.5 lb/hr

Comparable Documents

Powder which meets the requirements specified herein is qualified under the following specifications.

General Electric Company	B50TF33, Class A
Pratt & Whitney Aircraft	PWA 1321
Rolls-Royce Ltd.	MSRR 9507/4

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:31Z
Check the source to verify that this is the current version before use.